Chapter 2: Wireless Transmission and Medium Access Control

Comprehensive Study Guide — MOBAN 2025-2026

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1. Modulation

1.1 What is Modulation?

The transmission chain for digital data has **two modulation steps**:

```
Digital Data
    ↓  [Digital Modulation]
Analog Baseband Signal
    ↓  [Analog Modulation]
Radio Signal (at carrier frequency fc)
    ↓  [Transmitted over air]
Radio Signal (received)
    ↓  [Analog Demodulation]
Analog Baseband Signal
    ↓  [Synchronization & Decision]
Recovered Digital Data
```

- **Digital modulation**: Converts bits → analog baseband signal (ASK, FSK, PSK, QAM, OFDM)
- **Analog modulation**: Shifts baseband signal up to the radio carrier frequency (AM, FM, PM)

1.2 Radio Frequency vs. Wavelength

Frequency	Wavelength	Band	Name
30–300 Hz	10^4–10^3 km	ELF	Extremely Low Frequency
300–3000 Hz	10^3–10^2 km	VF	Voice Frequency
3–30 kHz	1000–10 km	VLF	Very Low Frequency
30–300 kHz	10–1 km	LF	Low Frequency
300–3000 kHz	1–0.1 km	MF	Medium Frequency
3–30 MHz	100–10 m	HF	High Frequency
30–300 MHz	10–1 m	VHF	Very High Frequency
300–3000 MHz	100–10 cm	UHF	Ultra-High Frequency
3–30 GHz	10–1 cm	SHF	Super-High Frequency

Frequency	Wavelength	Band	Name
30–300 GHz	10–1 mm	EHF	Extremely High Frequency (mm-waves)

Formula: $\lambda = c / f$, where $c = 3\times10^8$ m/s

> Higher frequency → shorter wavelength → shorter range → more free-space path loss

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1.3 Signal Propagation Ranges

Three concentric zones around a transmitter:

Zone	What happens
Transmission range	Data can be decoded at low error rate → communication possible
Detection range	Signal is detectable above noise floor, but error rate too high → no communication, PHY header may still be detected
Interference range	Signal cannot be detected but adds to background noise for others

> In real life, these zones are **not perfect circles** — obstacles, reflections, and time-varying conditions distort them.

1.4 Digital Modulation: ASK, FSK, PSK

Scheme	How it works	Bandwidth	Noise immunity	Use case
ASK (Amplitude Shift Keying)	Vary amplitude for 0/1	Low (= baud rate)	Poor (amplitude disturbed by fading)	Rarely used for wireless

Scheme	How it works	Bandwidth	Noise immunity	Use case
FSK (Frequency Shift Keying)	Different frequencies for 0/1	Larger (baud rate + carrier separation)	Good	Bluetooth (FHSS)
PSK (Phase Shift Keying)	Phase shift represents bits	Same as ASK	Better than FSK	WLAN, cellular

Why ASK is rarely used over wireless: Signal amplitude is constantly affected by fading, reflection, and distance. A constant amplitude cannot be guaranteed.

1.5 Bit Rate vs. Symbol Rate vs. Bandwidth

Key definitions:

- **Bit rate** (bps): number of bits transmitted per second
- **Symbol rate / Baud rate** (symbols/s or Baud): number of symbol changes per second
- **Bandwidth** (Hz): frequency range occupied by the signal

Key relationships:

```
Bit rate = Symbol rate × bits per symbol
bits per symbol = log₂(number of distinct symbols)

Minimum bandwidth:
  ASK / PSK:  BW_min = Symbol rate (Baud rate)
  FSK:        BW_min = Symbol rate + carrier separation
  OFDM:       Subcarrier spacing = 1 / symbol duration
```

Example — 4-level ASK (2 bits/symbol):

- 5 symbols/s → bit rate = 5 × 2 = 10 bps
- BW_min = 5 Hz (around carrier fc)

1.6 Advanced PSK

Variant	Bits/ symbol	IQ points	Notes
BPSK	1	2 (180° apart)	Most robust; used in satellite, poor-SNR links
QPSK	2	4 (90° apart)	Very common; balance of spectral efficiency and robustness
DQPSK	2	4	Differential variant — no reference signal needed at receiver
8-PSK	3	8	More bits/symbol but needs better SNR

- QPSK uses **IQ representation**: I axis (in-phase, 0°), Q axis (quadrature, 90°)
- Higher SNR → more phase points → more bits per symbol
- When SNR drops, constellations blur and bit errors rise

1.7 Quadrature Amplitude Modulation (QAM)

- Combines **amplitude AND phase** variation
- **n bits/symbol → 2^n discrete constellation points**
- n = 2 is identical to QPSK

Modulation	Bits/symbol	Constellation points	SNR requirement
BPSK	1	2	Lowest
QPSK	2	4	Low
16-QAM	4	16	Medium
64-QAM	6	64	High
256-QAM	8	256	Very High

Modulation	Bits/symbol	Constellation points	SNR requirement
1024-QAM	10	1024	Extreme

Used in: IEEE 802.11 (WLAN), LTE/5G (adaptive modulation — switch QAM order based on channel quality)

EVM (Error Vector Magnitude): Measures how accurately symbols are placed in the constellation. More negative (in dB) = more accurate.

1.8 Hierarchical Modulation

- Embeds a high-priority (HP) stream inside a lower-priority (LP) stream
- **Example (DVB-T)**: 64-QAM contains an embedded QPSK
  - 2 most significant bits = QPSK (HP — standard resolution TV)
  - Remaining 4 bits = additional QAM levels (LP — high resolution)
  - Under poor reception: receiver decodes only QPSK → still gets standard TV

1.9 OFDM (Orthogonal Frequency Division Multiplexing)

Concept: Split a high-rate data stream into many parallel low-rate streams, each on its own subcarrier.

```
High bit-rate stream
    ↓
Split into n substreams
    ↓
Each substream modulates its own subcarrier (at different frequency)
    ↓
All subcarriers transmitted simultaneously
```

Key properties:

- Subcarriers are **orthogonal** — the peak of one subcarrier aligns with the zero crossings of all others → no inter-carrier interference

- Implemented efficiently using **FFT/IFFT**
- **Longer symbol duration** → less sensitive to Inter-Symbol Interference (ISI)
- **Frequency-selective fading** only affects some subcarriers, not the whole signal

Used in: IEEE 802.11a/g/n/ac/ax (Wi-Fi), LTE, 5G NR, DAB

Subcarrier spacing formula: $\Delta f = 1 / T_{symbol}$

Example (IEEE 802.11a/g): 48 data subcarriers + 4 pilot subcarriers

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1.10 Pulse Shaping & Bandwidth

- **Narrow pulses (short symbol duration T)** → wide bandwidth ($BW \approx 1/T$)
- **Wide pulses (long symbol duration T)** → narrow bandwidth
- **Pulse shaping** (Raised Cosine filter): tradeoff between time-domain spreading and frequency containment
- ISI occurs when symbol duration < delay spread of the channel

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1.11 IQ Representation

- All RF modulation can be described in the **IQ plane** (complex plane)
- **I (In-phase)**: projection on x-axis (0° carrier)
- **Q (Quadrature)**: projection on y-axis (90° carrier)
- Each modulation scheme creates a characteristic **constellation diagram** (scatter plot)
- At the receiver, the scatter plot blurs with decreasing SNR until symbols can no longer be separated

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2. Spread Spectrum

2.1 Why Spread Spectrum?

Problem: Narrowband signals can be completely wiped out by narrowband interference or frequency-selective fading.

Solution: Spread the narrowband signal over a much wider bandwidth using a special code.

```
Narrow signal → [Spreading with code] → Wide broadband signal → [Transmit]
                                                                    ↓
                                                              Receiver
Wide signal + interference → [Despreading with same code] → Narrow signal recovered
                                                           + interference spread out
                                                           → [Bandpass filter] → Clean signal
```

2.2 Advantages & Disadvantages of Spread Spectrum

Advantages	Disadvantages
Protection against narrowband interference	Requires larger frequency band
Coexistence of multiple signals without coordination	Increased receiver complexity
Security / tap-proof (signal looks like noise)	Raises background noise level for others
Resistance to frequency-selective fading	(Requires precise power control for CDMA)

2.3 DSSS — Direct Sequence Spread Spectrum

Mechanism: XOR the data bit stream with a high-speed pseudo-random **chipping sequence**

```
User bit:       0  (duration = tb)
Chipping seq:   0 1 1 0 1 0 1  (each chip has duration tc)
Result (XOR):   0 1 1 0 1 0 1  (complement for bit=1)
```

Key parameters:

- **Spreading factor** s = tb / tc (number of chips per bit)
- **Bandwidth after spreading** = s × original bandwidth
- Civil applications: s = 10–100; Military: up to 10,000
- **Barker code** (11-chip: 10110111000) used in IEEE 802.11 DSSS

Advantages of DSSS:

- Reduces frequency-selective fading
- In cellular: adjacent base stations can use the **same frequency** (reuse = 1)
- **Soft handover**: multiple base stations can receive and combine the same signal

Disadvantages:

- Requires **precise power control** (near-far problem)
- Complex receiver (correlator, integrator, decision unit)

DSSS Receiver operation:

1. Demodulate carrier → spread baseband signal
2. Multiply by synchronized chipping sequence (correlator)
3. Integrate over one bit period
4. Decision unit: if sum > threshold → bit = 1, else bit = 0

2.4 FHSS — Frequency Hopping Spread Spectrum

Mechanism: Transmitter hops between frequencies according to a pseudo-random **hopping sequence**

Version	Definition	Properties
Slow hopping	Multiple bits per frequency hop (dwell time > bit period)	Simpler, cheaper, less robust
Fast hopping	Multiple hops per bit (dwell time < bit period)	More complex, much more robust to narrowband interference

Parameters:

- **Dwell time** (td): time spent on each frequency
- **Hopping sequence**: pseudo-random order of frequencies used

Examples:

- **Bluetooth**: 1600 hops/s, 79 channels, 1 MHz spacing, 2.4 GHz ISM band → slow hopping FHSS
- **GSM**: optional slow frequency hopping

Advantages vs DSSS:

- Simpler implementation
- Uses only a small portion of spectrum at any time

Disadvantages vs DSSS:

- Less robust to fading and multipath

2.5 DSSS vs FHSS Comparison

Feature	DSSS	FHSS
Mechanism	XOR with chipping sequence	Hop between frequencies
Interference resistance	Very high (spreads interference)	High (only short dwell time per frequency)
Complexity	High	Lower

Feature	DSSS	FHSS
Multipath robustness	Higher	Lower
Uses full bandwidth	Always	No (only one sub-band at a time)
Example	IEEE 802.11 (legacy), UMTS/CDMA	Bluetooth, GSM slow hopping

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3. Multiplexing & Multiple Access

3.1 Key Distinction

Term	Meaning
Multiplexing	Technical mechanism for sharing the medium (physical layer)
Multiple Access	Control/assignment protocol deciding who gets what resource

Four dimensions for multiplexing:

Dimension	Multiplexing	Multiple Access
Space (s)	SDM — Space Division Multiplexing	SDMA
Frequency (f)	FDM — Frequency Division Multiplexing	FDMA
Time (t)	TDM — Time Division Multiplexing	TDMA
Code (c)	CDM — Code Division Multiplexing	CDMA

> **Guard spaces are always needed** between channels (in time, frequency, space, or code) to avoid interference.

3.2 SDM / SDMA — Space Division

Concept: Separate channels by geographic separation. Same frequency can be reused in non-overlapping areas.

Key insight: Signal attenuation $\propto d^{-2}$ to d^{-4} → at sufficient distance, same frequency can be reused.

Cellular network example:

- Cells are grouped into **clusters** (3-cell, 7-cell, etc.)
- No two adjacent cells use the same frequency
- **Cell sectoring**: split one cell into 3 sectors using directional antennas → triple the frequency reuse

Cluster size	Frequency reuse	Co-channel interference
3-cell	High (aggressive reuse)	More
7-cell	Lower	Less

3.3 FDM / FDMA — Frequency Division

Concept: Divide the total spectrum into non-overlapping frequency bands. Each channel uses its band continuously.

```
f: [—guard—|—channel 1—|—guard—|—channel 2—|—guard—|—channel 3—|—guard—]
t: ←——————————————————— continuous ————————————————————————————————————→
```

Advantages:

- No dynamic coordination needed
- Works for both analog and digital signals
- Simple receivers (just tune to the right frequency)

Disadvantages:

- Wasteful when traffic is bursty (frequency reserved even when idle)
- Inflexible — fixed assignment
- Guard bands waste capacity

FDMA example (cellular): Each mobile user gets a dedicated uplink and downlink frequency pair.

3.4 TDM / TDMA — Time Division

Concept: All channels share the full spectrum, but each uses it during a different time slot.

```
f: ←─────────────── full spectrum ──────────────────→
t: [guard|slot 1|guard|slot 2|guard|slot 3|guard|slot 4|...]
          ↑ user A          ↑ user B
```

Advantages:

- Only one carrier at a time → no adjacent carrier interference
- Flexible: assign more slots to heavy users
- High throughput even with many users

Disadvantages:

- Requires **precise synchronization** between all senders
- Guard times waste capacity

TDMA in practice: GSM uses FDMA + TDMA (8 time slots per frequency channel)

3.5 CDM / CDMA — Code Division

Concept: All channels use the **same frequency at the same time**, separated by unique orthogonal codes.

```
f: ←—— same band ——→
t: ←—— same time ——→
c: [code A] [code B] [code C] ...  ← ALL SIMULTANEOUS
```

Implementation: Via spread spectrum (DSSS)

- Each user's data XORed with a unique chipping code
- At receiver: multiply received signal by known code → integrator → decision
- Orthogonal codes ensure no cross-user interference when all signals have equal power

Advantages:

- Excellent interference protection and tap-proof security
- Bandwidth efficient (all users share same band simultaneously)
- Soft handover in cellular networks

Disadvantages:

- **Near-far problem**: nearby strong signals drown out distant weak signals → precise power control is essential
- Complex receiver
- Lower per-user data rate (due to spreading)

Party analogy: CDM is like people speaking different languages in the same room. If you know the language (code), you understand the conversation; others hear it as noise.

3.6 Time + Frequency Multiplexing Combined

Concept: Each channel gets a specific frequency band for a specific time duration.

```
f: [f1][f2][f3]...
t:  1   2   3  ...
```

Advantages over pure CDM: Higher per-user data rates (no spreading overhead)
Examples: GSM (FDMA/TDMA/FDD), Bluetooth (FDMA/TDMA + FHSS)

3.7 Multiplexing Comparison Table

Method	Channel =	Advantages	Disadvantages	Examples
FDMA	Frequency band	Simple, no sync needed	Wasteful for bursty traffic, inflexible	GSM (combined with TDMA), FM radio
TDMA	Time slot	Flexible, high efficiency	Needs precise sync	GSM, DECT
CDMA	Code	Secure, efficient, soft handover	Near-far problem, complex, power control	UMTS (3G), CDMA2000
SDMA	Space/cell	Reuse spectrum geographically	Requires careful planning	All cellular systems
OFDMA	Frequency + Time subcarrier	Very flexible, robust to ISI	Complex synchronization	LTE, 5G NR, Wi-Fi 6

4. Duplex Techniques

4.1 Why Duplex?

Mobile communication is **bidirectional** (uplink: device→base, downlink: base→device). Two methods to support this:

Method	Mechanism	Notes
FDD (Frequency	Separate frequency bands for UL and DL	Simultaneous TX and RX possible; needs frequency guard band; e.g., GSM: 45 MHz

Method	Mechanism	Notes
Division Duplex)		separation
TDD (Time Division Duplex)	Alternate time slots for UL and DL on one frequency	Flexible UL/DL split; simpler hardware; but TX and RX cannot happen simultaneously

4.2 Half-duplex vs Full-duplex

Mode	Behavior	Examples
Half-duplex	Either transmit OR receive, not both simultaneously	Walkie-talkie, Wi-Fi (per channel), single-channel technologies
Full-duplex	Transmit AND receive simultaneously	Mobile phone with FDD, multi-channel technologies

> **Wi-Fi (802.11)** is half-duplex per channel — it can send OR receive but not both at the same time on the same channel. Wi-Fi 6 with multiple antennas enables effective full-duplex capability.

4.3 Common Combinations

Combination	Description	Example
FDMA/FDD	Each user has own UL+DL frequency pair	GSM (combined with TDMA)
FDMA/TDD	One frequency, UL and DL in alternating slots	DECT
FDMA/FDD/TDMA	Multiple users per frequency via time slots, FDD	GSM
FDMA/TDD/TDMA	Same but TDD	DECT

5. Medium Access Control (MAC)

5.1 Why Wireless Needs Special MAC

Can we reuse CSMA/CD from Ethernet (IEEE 802.3)? → NO

CSMA/CD assumption	Why it fails wirelessly
Signal strength uniform across wire	Strength ∝ d^{-2} to d^{-4} → colissions at receiver ≠ sender
Carrier sense detects all transmissions	**Hidden terminal problem**: sender can't sense a competing sender
Collision detection reliable	TX power >> RX power → sender can't hear collision at receiver

Three core MAC problems in wireless:

5.2 Hidden Terminal Problem

```
      range of A           range of C
  ←──────────────→     ←──────────────→
  A ──────────────── B ──────────────── C
```

1. **A senses channel** → idle → starts transmitting to B
2. **C senses channel** → can't hear A → thinks it's idle → starts transmitting to B
3. **Collision at B** — but neither A nor C detects it
4. **Result**: A is "hidden" from C and vice versa → wasted bandwidth, reduced throughput

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5.3 Exposed Terminal Problem

```
A —— B —— C —— D
```

1. B is sending to A
2. C wants to send to D (which is outside A's and B's range)
3. C senses carrier → detects B's signal → defers unnecessarily
4. C is "exposed" to B → wasted opportunity, reduced throughput

---

5.4 Near-Far Problem

```
A ———————————————————— C ——— B ——— C
```

- A and B both transmit to C at the same power
- B is much closer to C → B's signal drowns out A's signal at C
- **Critical for CDMA**: all signals must arrive at receiver with similar power
- **Solution**: precise power control (e.g., UMTS adjusts power 1500 times/second)

5.5 ALOHA

Context: Use TDM without central coordination — packet-based random access.

Variant	Mechanism	Max efficiency
Basic ALOHA	Send whenever you have data; retransmit after random backoff on collision	**18%**
Slotted ALOHA	Same but transmissions must start at slot boundaries (needs synchronization)	**36%**

How it works:

1. If you have data → send it
2. If collision detected → wait a random time → retransmit
3. Repeat until successful or max retries

Properties:

- No central coordination
- Works well under **light load**; breaks down under heavy load
- Cannot guarantee maximum delay or minimum throughput
- Still used in 3G/4G/5G for **initial random access** (random access channel)

5.6 MACA — Multiple Access with Collision Avoidance

Goal: Solve hidden terminal problem without a base station, using short signaling frames.

Two control packets:

- **RTS (Request To Send)**: Sender asks receiver for permission; contains sender address, receiver address, packet size/duration
- **CTS (Clear To Send)**: Receiver grants permission; contains same info

Procedure (solving hidden terminal):

```
A ────────────────── B ────────────────── C
      (A → B)                   (C → B: wants to send)

Step 1: A sends RTS to B (C cannot hear this)
Step 2: B sends CTS to A (C CAN hear this)
Step 3: C hears CTS → knows medium is reserved → defers for stated duration
Step 4: A transmits data to B → no collision
```

MACA also solves exposed terminal:

- B sends RTS to A
- C hears RTS, is NOT the destination → does not react
- A sends CTS (C cannot receive it — A is out of C's range)
- C correctly infers it can transmit (it missed CTS → A is outside range → no collision

possible there)

Limitation:

- RTS/CTS collisions can still occur (but short packets → low probability)
- RTS/CTS overhead is not negligible for small data packets
- MACA assumes symmetrical transmission conditions

IEEE 802.11 usage: MACA-style RTS/CTS is an optional mechanism in Wi-Fi (DCF mode).

5.7 MAC Scheme Summary

Scheme	Type	Solves	Limitations
CSMA/CD	Contention (wired)	—	Fails wirelessly
Basic ALOHA	Random access	No coordination needed	18% efficiency, no guarantees
Slotted ALOHA	Random access	Better than basic	36% efficiency, needs sync
MACA (RTS/CTS)	Random + reservation	Hidden & exposed terminal	RTS/CTS overhead
TDMA (fixed)	Scheduled	Guaranteed slots	Wasteful if slot unused
FDMA	Scheduled	No interference	Wasteful for bursty
CDMA	Spread-spectrum	Simultaneous access	Near-far, power control

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6. Key Formulas & Calculations

Modulation Formulas

Formula	Description
Bit rate = Symbol rate × $\log_2(M)$	M = number of distinct signal levels/symbols
Symbol rate = Bit rate / $\log_2(M)$	Inverse of above
BW_min (ASK/PSK) = Symbol rate	Minimum bandwidth equals baud rate
BW_min (FSK) = Symbol rate + f_separation	Add carrier separation
BW_min (FSK half-duplex) = Symbol rate + f_sep	Single band
BW_min (FSK full-duplex) = (Symbol rate + f_sep) / 2 per dir	Two equal halves
OFDM subcarrier spacing Δf = 1 / T_symbol	Orthogonality condition
OFDM bit rate = n_subcarriers × symbol_rate × bits_per_symbol	Total OFDM throughput
$\lambda = c / f$	Wavelength from frequency ($c = 3\times10^8$ m/s)
Spreading factor s = t_b / t_c	DSSS: bit period / chip period

Duplex / Bandwidth Formulas

Scenario	Bandwidth
Half-duplex	Full bandwidth used in one direction at a time
Full-duplex FDD	Total BW split equally: BW/2 per direction; carrier = center of each half

7. Assignment Solutions (Worked Examples)

Q1 — Symbol rate and bit rate from signal units

> An analog signal carries **6 bits per signal unit**. **1000 signal units/second** are sent. Find the symbol rate and the ASK bit rate.

- Symbol rate = **1000 symbols/s** (= 1000 signal units/s)
- Bit rate = Symbol rate × bits/symbol = 1000 × 6 = **6000 bps = 6 kbps**

Q2 — Symbol rate from bit rate and bits/symbol

> **Bit rate = 4 kbps**, each signal unit carries **8 bits**. Find the symbol rate.

- Symbol rate = Bit rate / bits per symbol = 4000 / 8 = **500 symbols/s**

Q3 — Minimum bandwidth for multi-level ASK (half-duplex)

> Find minimum bandwidth for a **4-level ASK** signal at **20 Mbps**, half-duplex.

- 4 levels → $\log_2(4)$ = **2 bits/symbol**
- Symbol rate = 20 Mbps / 2 = **10 M symbols/s**
- Minimum bandwidth = Symbol rate = **10 MHz**
- (Half-duplex: only one direction at a time → full 10 MHz used for that direction)

Q4 — Full-duplex FDD bandwidth and carrier frequencies

> Bandwidth = **10 MHz** (2.400 GHz to 2.410 GHz), full-duplex ASK. No gap between bands. Find carriers and bandwidth per direction.

- Total BW = 10 MHz, split equally → **5 MHz per direction**

- Direction 1: 2.400–2.405 GHz → carrier = 2.4025 GHz
- Direction 2: 2.405–2.410 GHz → carrier = 2.4075 GHz
- **Carrier frequencies: 2.4025 GHz and 2.4075 GHz**

Q5 — Minimum bandwidth for FSK (half-duplex)

> FSK signal at **2 Mbps**, half-duplex, carrier separation = **3 MHz**. Find spectral bandwidth.

- Symbol rate = 2 Mbps (1 bit/symbol for FSK)
- BW_min (FSK) = Symbol rate + carrier separation = 2 + 3 = **5 MHz**

Q6 — Maximum bit rate for FSK (full-duplex)

> Medium bandwidth = **12 MHz**, difference between carriers = **2 MHz**, full-duplex FSK.

- Full-duplex → BW per direction = 12 / 2 = **6 MHz**
- BW_FSK = Symbol rate + carrier separation
- 6 = Symbol rate + 2 → Symbol rate = **4 M symbols/s**
- FSK (binary): 1 bit/symbol → **Bit rate = 4 Mbps per direction**

Q7 — Symbol rate and bit rate for 8-PSK

> Bandwidth = **50 MHz**, 8-PSK signal.

- 8-PSK → $\log_2(8)$ = **3 bits/symbol**
- BW = Symbol rate → Symbol rate = **50 M symbols/s**
- Bit rate = 50 M × 3 = **150 Mbps**

Q8 — Bit rate for 32-QAM

> 32-QAM, symbol rate = **100 M symbols/s**.

- 32-QAM → $\log_2(32)$ = **5 bits/symbol**
- Bit rate = 100 M × 5 = **500 Mbps**

Q9 — OFDM bit rate and subcarrier spacing

> 64-QAM OFDM, **50 subcarriers**, symbol duration = **4 μs**.

- 64-QAM → $\log_2(64)$ = **6 bits/symbol**
- Symbol rate per subcarrier = 1 / 4 μs = **250 k symbols/s**
- Bit rate = 50 × 250k × 6 = 50 × 1.5 Mbps = **75 Mbps**
- Minimum subcarrier spacing Δf = 1 / T_symbol = 1 / 4 μs = **250 kHz**

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8. Flash Cards

Q: What are the two modulation steps in a radio transmitter?
A: (1) Digital modulation: converts bits to analog baseband signal. (2) Analog modulation: shifts baseband signal up to radio carrier frequency.

Q: Why is ASK rarely used for wireless transmission?
A: Signal amplitude is constantly disturbed by fading, reflections, and distance variations — a constant amplitude cannot be guaranteed wirelessly.

Q: What is the minimum bandwidth of an ASK signal?
A: Equal to the symbol rate (baud rate). BW_min = N_Baud.

Q: How does OFDM reduce Inter-Symbol Interference?
A: By splitting the high-rate stream into many low-rate parallel streams on different subcarriers, each with a much longer symbol duration. This makes the symbol period much longer than the channel's delay spread.

Q: What is the subcarrier spacing in OFDM?
A: Δf = 1 / T_symbol (orthogonality condition).

Q: What is the spreading factor in DSSS?

A: s = t_b / t_c (bit period / chip period). Bandwidth after spreading = s × original bandwidth.

Q: What is the key advantage of DSSS over FHSS?
A: DSSS is more robust against multipath fading and offers soft handover capability in cellular networks.

Q: What is slow vs. fast hopping in FHSS?
A: Slow hopping: multiple bits per frequency hop. Fast hopping: multiple frequency hops per bit (more robust, more complex).

Q: What are the 4 multiplexing dimensions?
A: Space (SDM), Frequency (FDM), Time (TDM), Code (CDM).

Q: What is the difference between multiplexing and multiple access?
A: Multiplexing is the physical mechanism for sharing the medium. Multiple access is the control protocol that assigns resources (time/frequency/code/space) to users.

Q: Why does CSMA/CD fail in wireless networks?
A: (1) Signal power decreases with distance so collisions at the receiver may not be detectable at the sender. (2) Hidden terminal problem: a sender may not be able to hear competing senders.

Q: Describe the hidden terminal problem.
A: Node A and C are both out of each other's range but both in range of B. A is transmitting to B; C cannot detect A's transmission, senses idle channel, transmits to B → collision at B that neither A nor C can detect.

Q: Describe the exposed terminal problem.
A: C can hear B transmitting to A, so C defers from transmitting to D. But C's transmission to D would not have caused a collision at A (A is out of C's range). C is unnecessarily blocked — wasted capacity.

Q: Describe the near-far problem.
A: A strong nearby transmitter drowns out a weaker far transmitter at the receiver. Critical for CDMA: without power control, the nearest device monopolizes the channel. Solution: closed-loop power control.

Q: What is the maximum efficiency of Basic ALOHA and Slotted ALOHA?
A: Basic ALOHA: 18%. Slotted ALOHA: 36% (double, because slotting halves the collision

window).

Q: How does MACA solve the hidden terminal problem?
A: Sender A sends a short RTS to receiver B. B responds with CTS which is heard by hidden node C. C defers for the duration indicated in the CTS, preventing it from transmitting during A's data exchange with B.

Q: What is FDD vs TDD?
A: FDD (Frequency Division Duplex): separate frequency bands for uplink and downlink — simultaneous TX/RX possible. TDD (Time Division Duplex): same frequency, alternating time slots for UL and DL — cannot TX and RX simultaneously.

Q: What is EVM?
A: Error Vector Magnitude — a measure of how accurately a transmitter places symbols on the constellation diagram. More negative dB = more accurate (better quality transmitter).

Q: What does hierarchical modulation do?
A: Embeds a high-priority (HP) signal in the most significant bits of a higher-order QAM, and low-priority (LP) data in the remaining bits. Poor-reception receivers can still decode the HP stream (e.g., standard TV resolution) even if they can't decode the full QAM.

Q: Which modulation scheme is most robust, and why?
A: BPSK — only 2 constellation points 180° apart, maximum distance between points → highest noise tolerance. Used in satellite links, deep coverage scenarios.

Q: What is the difference between CDM and CDMA?
A: CDM (Code Division Multiplexing) is the mechanism (all users share same frequency/time, separated by code). CDMA (Code Division Multiple Access) is the access control protocol that assigns codes to users.

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9. Exam Checklist

Modulation

- [ ] Draw and explain the full TX/RX chain (digital modulation → analog modulation →

transmission → demodulation → decision)

- [ ] Name and compare the 3 basic digital modulation schemes (ASK, FSK, PSK) on: bandwidth, noise immunity, complexity
- [ ] Calculate bit rate, symbol rate, and minimum bandwidth for ASK, FSK, PSK, QAM
- [ ] Explain QPSK using the IQ diagram (4 points, 2 bits/symbol)
- [ ] Explain QAM: formula for bits/symbol, constellation size, when to use higher QAM order
- [ ] Explain OFDM: concept, ISI resistance, subcarrier spacing formula, use cases
- [ ] Explain hierarchical modulation with DVB-T example
- [ ] Explain the pulse shaping tradeoff (time domain vs. frequency domain)
- [ ] Know the RF frequency bands and the wavelength formula

Spread Spectrum

- [ ] Explain why spread spectrum is needed (frequency-selective fading, narrowband interference)
- [ ] Draw the 5-step spread/despread process (power spectral density diagram)
- [ ] Explain DSSS: XOR with chipping sequence, spreading factor formula, advantages (soft handover, frequency reuse = 1)
- [ ] Explain FHSS: slow vs. fast hopping, dwell time, Bluetooth as example
- [ ] Compare DSSS and FHSS on: robustness, complexity, bandwidth usage

Multiplexing & Multiple Access

- [ ] Name and describe all 4 multiplexing dimensions and their multiple access counterparts
- [ ] Explain SDM/SDMA and cell clustering (3-cell and 7-cell clusters)
- [ ] Explain FDM/FDMA: pros/cons, when wasteful
- [ ] Explain TDM/TDMA: pros/cons, synchronization requirement
- [ ] Explain CDM/CDMA: orthogonal codes, power control, near-far problem
- [ ] Explain OFDMA as a combination used in LTE/5G

Duplex

- [ ] Distinguish FDD vs TDD — when is each preferred?
- [ ] Distinguish half-duplex vs full-duplex — example of each
- [ ] Identify the duplex scheme used in GSM (FDD), DECT (TDD)

MAC

- [ ] Explain why CSMA/CD fails in wireless networks (3 reasons)
- [ ] Describe the hidden terminal problem with a diagram
- [ ] Describe the exposed terminal problem with a diagram
- [ ] Describe the near-far problem and its solution
- [ ] Explain Basic ALOHA and Slotted ALOHA — efficiency figures and assumptions
- [ ] Explain MACA: RTS/CTS exchange, how it solves hidden terminal, how it solves exposed terminal
- [ ] State the overhead/limitation of MACA

Calculations

- [ ] Solve symbol rate / bit rate / bandwidth problems for ASK, FSK, PSK, QAM
- [ ] Solve OFDM bit rate and subcarrier spacing problems
- [ ] Apply full-duplex vs half-duplex to bandwidth/carrier calculations

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Sources: Moerman, I. (2025). MOBAN Chapter 2 slides. Schiller, J. (2003). Mobile Communications, 2nd ed. Popovski, P. (2020). Wireless Connectivity.